

NETWORKING 05

DNS (DOMAIN NAME SYSTEM)

- 👉 Internet works with **numbers** (IP)
- 👉 Humans use **names** ([google.com](https://www.google.com))

📌 So:

DNS = translator (name → IP address)

DNS FLOW (VERY IMPORTANT)

Step-by-step:

- 1 Browser asks DNS
👉 "What is IP of [google.com](https://www.google.com)?"
- 2 DNS replies
👉 "Here is IP"
- 3 Browser connects
👉 Opens website

TASK (PRACTICAL)

Run:

```
nslookup google.com
```

WHERE TO RUN

- Linux / WSL → Terminal
- Windows → CMD
- Mac → Terminal

YOU WILL SEE

```
Name: google.com
Address: 142.250.xxx.xxx
```

- 👉 Write this IP

What if DNS is COMPROMISED?

👉 This is called:

DNS Spoofing / DNS Poisoning

⚠️ Example attack:

You type:

google.com

But DNS gives:

Fake IP → hacker server

➡ You open FAKE website 😬

➡ You enter password

➡ Hacker steals data

So:

DNS = very sensitive target for attackers

WHY DNS IMPORTANT IN CYBER ATTACKS

1 Redirection attack

- User → fake website

2 Malware communication

- 👉 Virus inside system contacts:

malicious-domain.com

- DNS resolves it
- Malware talks to hacker server

3 Data exfiltration

- 👉 Hackers send data using DNS queries
- Hard to detect